

FAIRterm 2.0: Towards FAIR Terminologies Resources for EOSC

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Abstract—This paper presents the roadmap from FAIRterm 2.0 to initiatives which outline a strategic shift toward the integration of terminological and lexicographical resources in compliance with FAIR principles. While FAIRterm 2.0 currently operates within XML-based ISO standards dedicated to terminology management, it sets the foundation for future interoperability with Linguistic Linked Open Data models such as OntoLex-Lemon. We highlight ongoing efforts to bridge terminological practices and Semantic Web standards, including prototype tools for TBX-to-OntoLex translation. This work responds to emerging demands for multilingual, interoperable, and semantically rich knowledge infrastructure within EOSC, building upon recent research advocating for the convergence of lexical and terminological systems.

Index Terms—FAIR Terminology, Terminological Interoperability, EOSC Integration

I. INTRODUCTION

Correct terminology plays a fundamental role in supporting research by promoting clarity, precision, and effective communication: it ensures that complex concepts are conveyed accurately by reducing ambiguity and fostering a shared vocabulary among researchers; it is particularly crucial in interdisciplinary and multilingual environments, where ambiguity can make collaboration and knowledge transfer difficult. Therefore, terminologies underpin the semantic layer of research data. However, existing terminological repositories suffer from fragmentation, lack of machine-actionability, and limited openness [1]. For instance, the European Union's flagship termbase, IATE [2], one of the largest terminology databases ever built, is primarily managed as a top-down institutional database where researchers have no direct mechanism to contribute terms, align them with research outputs, or integrate domain-specific updates. This limits the responsiveness and curation of terminology resources to both emerging and established scientific domains. Without FAIR terminologies, interoperability across research data, policies, and disciplines remains fragile, particularly in cross-domain or multilingual settings [3]. At the same time, the growing reliance on AI-driven tools – especially Large Language Models (LLMs) and machine translation systems – highlights the need for high-quality, domain-specific terminologies to ensure accurate interpretation, multilingual consistency, and semantic disambiguation [4]. LLMs often hallucinate or misapply specialized terms

in domain-specific contexts due to ambiguous, incomplete, or inconsistent terminological data.

In response to these challenges, the *FAIR terminology* approach has emerged [5]. It applies the FAIR Principles [6] - Findability, Accessibility, Interoperability, and Reusability - to terminology resources, ensuring that terms can be openly accessed, semantically aligned, and linked to authoritative sources across languages and domains. FAIR terminologies are not only machine-readable but also versioned, citable, and embedded in scholarly communication workflows, thus enabling both automated reasoning and human interpretation. The FAIR terminology paradigm relies upon the complementary application of the three ISO TC/37 SC3 standards: ISO 16642: 2017 [7]; ISO 12620: 2019 [8], and ISO 30042: 2019 [9] and represent the basic building blocks to be used for a multilingual FAIR terminology resource. These standards remain the only internationally recognized and FAIR-compatible framework for terminology management in which TBX (Terminology Base eXchange), based on XML, is the only one specifically designed for the exchange of terminological data.

In this work, we have two main objectives:

- First, we present the new release of the *FAIRterm Web Application 2.0*, detailing its concept-oriented structure, multilingual features, and compliance with ISO terminology standards, and contrasts it with the previous version to highlight key advancements in usability, functionality, and data modeling.
- Second, we outline a proposed framework, *TERM4EOSC (Terminology for EOSC Repositories)*, to enhance the machine-actionability and infrastructural integration of FAIRterm 2.0. TERM4EOSC envisions the alignment of terminological data with persistent identifiers, open metadata standards, and EOSC services, thus laying the groundwork for a scalable, collaborative, and FAIR-compliant terminology infrastructure for European research communities.

The remainder of this paper is organized as follows. Section II reviews the evolution of FAIR terminology practices, contrasting the initial FAIRterm 1.0 prototype with the more advanced and modular architecture of FAIRterm 2.0. Section III introduces the TERM4EOSC initiative as a forward-

looking framework to align terminology resources with Semantic Web standards. In particular, we discuss the limitations of TBX for full FAIRification, ongoing efforts to convert terminological data into the Linguistic Linked Open Data ecosystem, and early tools enabling translation to OntoLex-Lemon. Finally, Section IV offers concluding reflections on the broader implications for open science and outlines future directions for advancing interoperable, FAIR-aligned terminology services across EOSC.

II. FAIR TERMINOLOGY: PRINCIPLES AND EVOLUTION

The notion of FAIR terminology has emerged from efforts to systematize the representation, curation, and sharing of domain-specific terminologies in accordance with the FAIR principles. A foundational milestone in this direction is the TRIMED project [10]–[12], which developed a multilingual terminology resource in the medical domain to enable integration across datasets, clinical documentation, and communication between patients and physicians. TRIMED provided early evidence of how principles such as interoperability and reusability can be embedded in terminological entries to support transparent knowledge exchange.

Building on this foundation, the broader conceptual framework of FAIR terminology was described in [5]. This book is completely dedicated to the approach and methodology for the organization of data and implementation of terminological databases using the available standards on terminology. This framework argues for the transformation of terminologies into FAIR digital objects which should not only be curated in a systematic, multilingual, and domain-sensitive manner, but also published with persistent identifiers, machine-readable metadata, and explicit links to the research outputs that define, use, or revise the terms. Subsequent works showcased the implementation of this paradigm in two different fields like international disarmament and commercial products like wine, glass, and textile [13], [14].

In [15], the FAIR terminology approach was used to design and implementation of a web application aimed at facilitating access to a multilingual terminology database created by the Translators without Borders (TWB/CLEAR Global) [16]. The database contains essential concepts translated into 39 languages, for a total of 9,506 term and it is specifically created for speakers of marginalized languages that often cannot access information they need and want in a language and format they understand.

The FAIRterm Web Application (FWA) [17], an initiative carried out by the University of Padova, was designed and implemented to operationalize the FAIR terminology approach into a Web platform for collaborative terminology management. The initial version focused on supporting the manual creation and publication of terminological entries according to FAIR-aware templates. In 2021, it was one of the documented use cases in the EOSC Task Force on FAIR Metrics and Data Quality Charter [18]. In 2021–2022, the FWA was selected by the Terminology Coordination Unit of the European Parliament as the platform to support the

Terminology Without Borders (YourTerm) initiative [19], [20], facilitating multilingual collaboration among universities and institutional partners [21].

FWA has been deployed successfully and used by more than 800 users across 30 countries and 80 universities and institutions for multiple use cases and specialized domains (such as medicine, law, economics, culture, etc.).

From a technical viewpoint, FWA is implemented as an R Shiny web application [22]. It leverages Shiny modules for UI structuring and reactive programming to handle user inputs dynamically. Data is managed through R tabular and stored in a MySQL backend. The application also includes APIs for exporting entries in the standard TBX/XML format, enabling interoperability with external services.

A. FAIRterm 1.0

FAIRterm 1.0 is the results of a collaboration between the Department of Linguistic and Literary Studies and the Department of Information Engineering at the University of Padova, later formalized with the first Italian Center of Studies on Computational Terminology (CENTRICO) [23]. It was the first documented web application designed to align with both the FAIR principles and the ISO TC 37/SC 3 standards for terminology management, including ISO 16642:2017, ISO 12620:2019, and ISO 30042:2019 [24].

This tool derives from the implementation of the TriMED database [10] and supports the compilation of multilingual terminological entries by providing a user-friendly interface shown in Figure 1. FAIRterm 1.0 was designed to support translation-oriented terminology work. For this reason, it adopts a term-based approach rather than a concept-oriented one. The interface is structured around a twofold view that distinguishes between *source* and *target* terms, reflecting the typical workflow of translators and language professionals. Although the underlying terminological entry is not inherently directional and does not privilege a source or target language, this layout facilitates bilingual or multilingual term alignment and ensures clarity during the compilation of records across different languages. On the other hand, the main data structure is still concept-oriented, adhering to the TBX format to ensure interoperability and standardization. This dual approach enables effective terminology management and supports collaborative projects, such as those within the “Terminology Without Borders” initiative, enhancing the quality and consistency of terminology resources across different languages and domains.

In order to compare FAIRterm version 1.0 with version 2.0 (presented in the following section), we highlight the main steps and how the user can interact with these elements.

- 1. Select project and subproject:** Since terminology studies terms and concepts of a specialized domain, the first step is to declare the context for the term entry. In this example, the user chooses a “main project” (e.g., TECH) and a relevant “subproject” (e.g., Natural Language Processing) from the dropdown menus on the left panel.

Fig. 1. Screenshot of the FAIRterm Web Application 1.0. This version was used in the context of the Terminology Without Borders (YourTerm) initiative of the Terminology Coordination Unit of the European Parliament.

- 2a. **Set the source language:** In order to search the terms in the database, the user must select at least a source language. In the example, select the language of the original term (e.g., English) from the dropdown list labeled “Source language”.
- 2b. **Set the target language:** Optionally, the user can select the target language in order to activate the right side of the entry and the corresponding data. In the example, select the language into which the term is being translated or aligned (e.g., French) from the “Target language” dropdown.
3. **Search a term:** When languages are chosen, the user can search for a term (in the source language). In the example, use the “Search a term” field to locate an existing entry in the database. This supports quick access to previously recorded terms.
4. **Enter or edit the term:**
 - 4a. In the “Source Term” field, type or edit the term in the selected source language (e.g., *decision tree*).
 - 4b. In the “Target Term” field, type or edit the equivalent term in the target language (e.g., *arbre de décision*).
5. **Enter term entry info:** In the “Categories” section, the user can fill in or view detailed information of the entry divided into four categories: Formal features, Semantics, Variation, Usage.

6. **Save or export the record:** The user can download the record in tabular format or in the standard TBX format which can be later integrated in other tools.

B. FAIRterm 2.0: A New Concept-Oriented Platform for Multilingual Terminology

FAIRterm 2.0 represents a substantial evolution of the platform in terms of both conceptual modeling and technical implementation. The new version adopts a concept-oriented structure in accordance with ISO TC 37/SC 3 standards. Terminological entries are now organized hierarchically in a three-layer structure also from a visual point of view (see Figure 2):

- by concept, then
- by language section, and finally
- by term section

allowing for greater clarity in multilingual representation and semantic disambiguation. One of the major advancements is the clear separation between the concept level—non-linguistic and abstract—and the linguistic level, which includes terms and natural language definitions associated with specific languages. Concepts are explicitly modeled and uniquely identified through non-linguistic identifiers, allowing for the unambiguous representation of domain-specific knowledge. This facilitates the encoding of hierarchical relationships such as superordinate and subordinate links among concepts, thus enabling a more robust and semantically rich organization

Fig. 2. Screenshot of the FAIRterm Web Application 2.0.

of terminological entries. Definitions are now systematically assigned at the language level, supporting multilingual clarity while avoiding redundancy across languages. Furthermore, the inclusion of multiple terms under the same concept in a single language section implicitly signals synonymy, providing users with a practical overview of lexical variation within conceptual boundaries.

FAIRterm 2.0 also integrates domain classification by aligning with *EuroVoc* [25], the European Union’s multilingual and multidisciplinary thesaurus. Each concept is associated with one or more standardized EuroVoc categories, facilitating thematic consistency and interoperability with EU institutions and international repositories. These enhancements improve both human and machine readability and support the use of FAIRterm 2.0 in knowledge-based applications, such as linked data environments, domain-specific ontologies, and multilingual information systems.

In order to compare the two versions, we used the same steps (even though they are not always necessarily identical) to help the reader to visually focus on the different structure. We also used exactly the same example with the term “decision tree”.

1. **Select subject field and concept:** Like in version 1.0, the first step is the choice of the domain in which the user should start to work. Choose a subject field from the dropdown (e.g., *6411 – technology and technical regulations*) and then select or input the corresponding concept ID. This anchors the term pair within a structured

conceptual domain. Options to reset fields or add a new concept entry are also available. The additional metadata related to the concept level (subdomain, superordinate concept, etc.) are initially hidden and can be displayed using the “show/hide” button (the same for the language and the term level).

3. **Search:** Since concepts are extra-linguistic entities which are identified by numbers (or any permissible sequence of characters), if the user cannot remember them, there is a search bar at the top to locate specific terms that designate the concepts. In this way, we can retrieve the terminological entry connected to that term.
2. **Show/add language(s) (2a/2b):** When a concept is selected, FAIRterm 2.0 will automatically load all the languages associated to it, in the example only English and French. The user can add as many languages as the ones contained in the ISO 639: 2023 [26].
4. **Show/add terms (4a/4b):** Since there is no longer the idea of source/target languages, all the terms that designate the concept for each language are shown with separate sections.
5. **Enter term entry info:** For each term section, the user can fill in or view detailed information of the entry. In the example, the first metadata concerns the usage of the term (preferred, admitted, deprecated, obsolete).
6. **Save or export the record (not shown in the screenshot):** The user can download the record in tabular format or in the standard TBX format which can be later

integrated in other tools.

From a graphical standpoint, FAIRterm 2.0 introduces a cleaner and more modular interface. Unlike the fixed bilingual table in version 1.0, the new interface uses collapsible panels for each concept and language. This allows users to focus on specific components of the entry and better manage multilingual data. Input validation, metadata prompts, and help guides have also been improved to support non-expert users.

C. Current Usage Statistics

Launched in November 2024, as of May 2025, the FAIRterm 2.0 application has over 180 registered users, including students, researchers, and institutional collaborators. The database includes more than 1,600 concepts, 3,250 concept-language pairs, and 3,800 terms across 67 subject fields.

III. TOWARD FAIRTERM 2.0 IN TERM4EOSC

TERM4EOSC is a new initiative carried out by the center of studies in computational terminology (CENTRICO) at the University of Padova. Building on the experience and technical groundwork of the FAIRterm Web Application 2.0, it aims to integrate expert-curated terminologies into the European Open Science ecosystem by aligning them with FAIR principles and linking them with key EOSC services and infrastructure.

This evolution builds on the lessons learned from the original FAIRterm Web Application and addresses foundational limitations in the existing standards for termbase exchange, particularly the TBX format. In fact, despite TBX's status as an ISO standard widely adopted in the translation and terminology communities, recent studies have highlighted its shortcomings in terms of semantic interoperability, machine-actionability, and metadata expressivity—critical dimensions for full FAIR compliance [3]. These limitations constrain the integration of TBX-based resources into Linked Data ecosystems and limit their potential for reuse across disciplinary and infrastructural boundaries.

The initiative aims to advance this paradigm by serializing these standards-compliant resources into Semantic Web formats compatible with the Linguistic Linked Open Data (LLOD) cloud [27], such as RDF, OWL, and SKOS. In doing so, TERM4EOSC transforms static terminologies into FAIR Digital Objects enriched with persistent identifiers, machine-readable metadata, and explicit links to research outputs and infrastructure. This transition is anchored in semantic web technologies and ontology-driven modeling, with *OntoLex-Lemon* serving as a target framework for enhancing the expressiveness and interoperability of terminological resources [28], [29].

As a concrete step toward this transformation, TERM4EOSC is designing a conversion tool that maps TBX entries to *OntoLex-Lemon* structures, incorporating terminological, lexical, and concept-oriented elements. This process is informed by prior work on ontology-based terminology modeling which emphasizes the need for

preserving terminological rigorosity while supporting machine reasoning and semantic linking [30].

By integrating TBX content into the LLOD cloud, *FAIRterm 2.0* will serve as a pivotal tool in the ecosystem of EOSC-oriented language resources, bridging legacy formats and future-proof FAIR terminologies through semantic technologies and community-aligned standards. *FAIRterm 2.0* will empower users to query, link, and enrich terminologies in multilingual, multidisciplinary, and infrastructure-aligned contexts.

IV. CONCLUDING REMARKS

The advancement of FAIR principles in the field of terminology marks an important step in bridging linguistic data with the broader goals of open science. As terminological resources continue to play a foundational role in scientific communication, regulatory harmonization, and semantic interoperability, the need for robust, machine-actionable, and interoperable formats becomes increasingly urgent. Traditional standards like TBX, while valuable, fall short in expressing rich semantic relationships and integrating seamlessly with the evolving Linked Open Data landscape.

FAIRterm 2.0 represents an important foundational step in the transformation of terminological data toward full FAIR compliance. While it does not yet adopt Semantic Web technologies such as *OntoLex-Lemon*, it establishes the technical and conceptual groundwork necessary for such a transition. By improving the structure, accessibility, and standardization of terminological resources, FAIRterm 2.0 prepares the way for the more advanced, linked data-driven approach envisioned in the TERM4EOSC initiative. This progression in phases supports the gradual adoption of interoperable, machine-readable formats that are essential for building a transparent, multilingual, and cross-disciplinary terminological infrastructure. TERM4EOSC positions this evolution within the broader European Open Science Cloud (EOSC) ecosystem. By focusing on the harmonization and FAIRification of terminology resources, it fosters cross-domain interoperability and supports EOSC's mission to make research outputs more accessible and reusable. Through strategic integration with platforms such as IATE, CLARIN, and OpenAIRE, TERM4EOSC will offer a bridge between domain-specific terminologies and the federated data infrastructure that underpin European science.

In particular, TERM4EOSC aims to deliver:

- A machine-actionable terminology model based on open and interoperable standards;
- open-source tools for editing, linking, and publishing terminologies as FAIR Digital Objects;
- Integration pathways for connecting terminologies to EOSC platforms and data pipelines;
- licensing and persistence strategies for sustainable reuse and attribution.

As a final remark, the direction of this work aligns with recent scholarship advocating for a closer integration between terminological databases and lexical resources, particularly in multilingual and interdisciplinary contexts. [31] argue for a conceptual and technological convergence of terminological

and lexical approaches to better reflect the dynamic, usage-based nature of knowledge organization. The FAIRterm 2.0 roadmap and the TERM4EOSC initiative respond to this call by supporting formats and workflows that allow terminological entries to be enriched with lexical features, ultimately enhancing reusability, interoperability, and semantic depth in both scholarly and infrastructural settings.

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